

EXPERIMENT-8: Session tracking with cookies and HTTP sessions.

Explore session tracking mechanisms like cookies and HTTP sessions to maintain user transaction history.

Objective : Implement persistence of user state across requests using session techniques.

Tools : Cookies, HTTP Session, server-side state.

Outcome : A web application that preserves user transaction history using sessions and cookies.

Answer:

Use cookies or session storage to keep the user's cart and login state across page visits.

📁 Project Structure

SessionTrackingApp/

├─ src/com/lab/session/

| ├─ LoginServlet.java -> login form, sets cookies, creates session

| ├─ CartServlet.java -> add items, shows transaction history (session)

| └─ LogoutServlet.java -> invalidates session, cookies remain

└─ WebContent/

├─ index.html

└─ WEB-INF/web.xml

Prerequisites

- JDK 8+

- Apache Tomcat 9 (`javax.servlet.\*`)

No database or JDBC driver needed for this lab.

Deploy

Option A: Eclipse

1. New → Dynamic Web Project → name it `SessionTrackingApp`.
2. Replace generated `src` / `WebContent` with the files here.
3. Run As → Run on Server (Tomcat 9).

File Name: CartServlet.java

Code:

```
package com.lab.session;
```

```
import java.io.IOException;
```

```
import java.io.PrintWriter;
```

```
import java.text.SimpleDateFormat;
```

```
import java.util.ArrayList;
```

```

import java.util.Date;
import java.util.List;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 * CartServlet
 * -----
 * Uses HttpSession (server-side state) to keep a running transaction
 * history for the logged-in user, across multiple requests, without
 * touching a database. The cart is lost when the session ends
 * (logout, timeout, or server restart) — unlike cookie data.
 */
@WebServlet("/cart")
@SuppressWarnings("unchecked")
public class CartServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false); // false = don't create a new one
        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login");
            return;
        }

        render(request, response, session, null);
    }

```

@Override

protected void doPost(HttpServletRequest request, HttpServletResponse response)

throws ServletException, IOException {

HttpSession session = request.getSession(false);

if (session == null || session.getAttribute("username") == null) {

response.sendRedirect("login");

return;

}

String item = request.getParameter("item");

String priceStr = request.getParameter("price");

try {

double price = Double.parseDouble(priceStr);

List<String> cart = (List<String>) session.getAttribute("cart");

if (cart == null) {

cart = new ArrayList<>();

}

String timestamp = new SimpleDateFormat("HH:mm:ss").format(new Date());

cart.add(LoginServlet.escape(item) + "|" + price + "|" + timestamp);

session.setAttribute("cart", cart); // updated transaction history

response.sendRedirect("cart"); // Post/Redirect/Get

} catch (NumberFormatException e) {

render(request, response, session, "Price must be a valid number.");

}

}

private void render(HttpServletRequest request, HttpServletResponse response,

HttpSession session, String error) throws IOException {

String username = (String) session.getAttribute("username");

List<String> cart = (List<String>) session.getAttribute("cart");

if (cart == null) cart = new ArrayList<>();

```

int visitCount = LoginServlet.getVisitCount(request);

response.setContentType("text/html;charset=UTF-8");
try (PrintWriter out = response.getWriter()) {
    out.println("<!DOCTYPE html><html><head><title>Your Cart</title>");
    out.println(LoginServlet.baseStyle());
    out.println("</head><body>");

    out.println("<h1>Welcome, " + LoginServlet.escape(username) + "</h1>");
    out.println("<p class='note'>Session ID: " + session.getId()
        + "<br>Session created: " + new Date(session.getCreationTime())
        + "<br>Session will expire after "
        + (session.getMaxInactiveInterval() / 60) + " minutes of inactivity."
        + "<br>Cookie-tracked total visits (persists beyond this session): " + visitCount
        + "</p>");

    if (error != null) {
        out.println("<p style='color:#b00020;font-weight:bold;'>" + error + "</p>");
    }

    out.println("<fieldset><legend>Add Item to Cart</legend>");
    out.println("<form method='post' action='cart'>");
    out.println("<label>Item</label><br><input type='text' name='item' required><br>");
    out.println("<label>Price</label><br><input type='number' step='0.01' name='price'
required><br>");
    out.println("<button type='submit'>Add to Cart</button>");
    out.println("</form></fieldset>");

    out.println("<h2>Transaction History (this session)</h2>");
    if (cart.isEmpty()) {
        out.println("<p>No items added yet.</p>");
    } else {
        out.println("<table><tr><th>#</th><th>Item</th><th>Price</th><th>Time
Added</th></tr>");
        double total = 0;

```

```

        int i = 1;
        for (String entry : cart) {
            String[] parts = entry.split("\\|");
            double price = Double.parseDouble(parts[1]);
            total += price;
            out.println("<tr><td>" + (i++) + "</td><td>" + parts[0] + "</td><td>$"
                + String.format("%.2f", price) + "</td><td>" + parts[2] + "</td></tr>");
        }
        out.println("</table>");
        out.println("<p><b>Total: $" + String.format("%.2f", total) + "</b></p>");
    }

    out.println("<p><a href='logout'>Logout</a></p>");
    out.println("</body></html>");
}
}
}

```

File Name : index.html

```

code: <!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Bookstore - Session Tracking Demo</title>
    <meta http-equiv="refresh" content="0; url=login">
</head>
<body>
    <p>Redirecting to <a href="login">login</a>...</p>
</body>
</html>

```

File name : LoginServlet.java

code:

```

package com.lab.session;

```

```

import java.io.IOException;
import java.io.PrintWriter;
import java.util.ArrayList;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 * LoginServlet
 * -----
 * Demonstrates COOKIES (client-side, survive across browser sessions)
 * versus HTTP SESSION (server-side, lasts only for the current session).
 *
 * - A "username" cookie remembers who last logged in (7-day expiry).
 * - A "visitCount" cookie counts logins over time, even across
 *   different HttpSession / browser restarts.
 * - On successful login, a fresh HttpSession is created to hold the
 *   shopping cart / transaction history for CartServlet.
 */
@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String rememberedUser = getCookieValue(request, "username");
        int visitCount = getVisitCount(request);

```

```

response.setContentType("text/html;charset=UTF-8");
try (PrintWriter out = response.getWriter()) {
    out.println("<!DOCTYPE html><html><head><title>Login</title>");
    out.println(baseStyle());
    out.println("</head><body>");
    out.println("<h1>Bookstore Login</h1>");

    if (rememberedUser != null) {
        out.println("<p class='note'>Cookie remembers you as <b>"
            + escape(rememberedUser) + "</b>. You have visited "
            + visitCount + " time(s) before (tracked via cookie, "
            + "not session).</p>");
    } else {
        out.println("<p class='note'>No 'username' cookie found yet — this looks like a new
browser/session.</p>");
    }

    out.println("<form method='post' action='login'>");
    out.println("<label>Username</label><br>");
    out.println("<input type='text' name='username' required value='"
        + (rememberedUser != null ? escape(rememberedUser) : "") + "'><br>");
    out.println("<button type='submit'>Login</button>");
    out.println("</form>");
    out.println("</body></html>");
}
}

```

@Override

```

protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    String username = request.getParameter("username");
    if (username == null || username.trim().isEmpty()) {
        response.sendRedirect("login");
        return;
    }
}

```

```

// ---- HTTP SESSION: create/replace session, store user + empty cart ----
HttpSession session = request.getSession(true); // true = create if not exists
session.setAttribute("username", username);
session.setAttribute("cart", new ArrayList<String>());
session.setMaxInactiveInterval(15 * 60); // session expires after 15 min idle

// ---- COOKIE 1: remember username for 7 days ----
Cookie userCookie = new Cookie("username", username);
userCookie.setMaxAge(7 * 24 * 60 * 60); // 7 days in seconds
userCookie.setPath("/");
response.addCookie(userCookie);

// ---- COOKIE 2: increment persistent visit counter ----
int visitCount = getVisitCount(request) + 1;
Cookie visitCookie = new Cookie("visitCount", String.valueOf(visitCount));
visitCookie.setMaxAge(30 * 24 * 60 * 60); // 30 days
visitCookie.setPath("/");
response.addCookie(visitCookie);

response.sendRedirect("cart");
}

// ----- helpers -----

static String getCookieValue(HttpServletRequest request, String name) {
    Cookie[] cookies = request.getCookies();
    if (cookies == null) return null;
    for (Cookie c : cookies) {
        if (c.getName().equals(name)) return c.getValue();
    }
    return null;
}

static int getVisitCount(HttpServletRequest request) {
    String v = getCookieValue(request, "visitCount");

```

```

try {
    return v != null ? Integer.parseInt(v) : 0;
} catch (NumberFormatException e) {
    return 0;
}
}

static String escape(String s) {
    if (s == null) return "";
    return s.replace("&", "&amp;").replace("<", "&lt;")
        .replace(">", "&gt;").replace("\"", "&quot;");
}

static String baseStyle() {
    return "<style>"
        + "body{font-family:Arial,sans-serif;max-width:600px;margin:50px auto;padding:0 15px;}"
        + "input,button{padding:6px;margin:4px 0;}"
        + "input[type=text]{width:95%;}"
        + ".note{background:#eef5ff;padding:10px;border-radius:4px;}"
        + "table{width:100%;border-collapse:collapse;margin-top:15px;}"
        + "th,td{border:1px solid #ccc;padding:8px;text-align:left;}"
        + "th{background:#f0f0f0;}"
        + "</style>";
}
}

```

File Name : LogoutServlet.java

code: package com.lab.session;

```

import java.io.IOException;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

```

```

import javax.servlet.http.HttpSession;

/**
 * LogoutServlet
 * -----
 * Ends the HttpSession (transaction history / cart is discarded).
 * The "username" and "visitCount" COOKIES are deliberately left alone,
 * so logging back in still remembers who you are and how many times
 * you've visited — proving cookies outlive the session.
 */
@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);
        if (session != null) {
            session.invalidate(); // destroys server-side session data (the cart)
        }
        response.sendRedirect("login");
    }
}

```

File Name: web.xml

```

code: <?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
        http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd"
    version="4.0">

    <display-name>SessionTrackingApp</display-name>

```

<!-- Servlets registered via @WebServlet annotations:

 LoginServlet -> /login

 CartServlet -> /cart

 LogoutServlet -> /logout -->

<welcome-file-list>

 <welcome-file>index.html</welcome-file>

</welcome-file-list>

<!-- Default session timeout in minutes (overridden per-session in LoginServlet) -->

<session-config>

 <session-timeout>15</session-timeout>

</session-config>

</web-app>